# BA820 – Project M4

## Cover Page

- **Project Title:** Hollywood Age Gap - Beyond the Timeline: Mapping the Evolution of Casting Norms and Romantic Age Gaps in Cinema
- **Section and Team Number:** B1 Team 10
- **Student Name:** Shanmathi Sivakumar

## 1. Refined Problem Statement & Focus

My question in this milestone stays focused on within film patterns: In movies that contain multiple romantic couples, do age gaps concentrate in one standout pairing or do they recur across several couples within the same film, indicating a consistent within film casting pattern rather than a single isolated relationship choice.

The domain value is that these two situations mean different things in practice. A film with one extreme couple can look like a one-off story choice, while repeated gaps across couples looks more like a consistent casting norm. This distinction matters for how we interpret representation trends and how we describe what a movie is showing audiences.

What changed in Milestone 4 is how I measure and represent this headline versus recurrence idea at the movie level. In earlier work, the movie features leaned mostly on the largest one or two gaps, such as the second largest gap and top gap dominance, plus a large gap cutoff. That approach captured whether a movie had a very large gap, but it did not fully capture whether the gap pattern was concentrated in one couple or spread across multiple couples. In this milestone, I refine the focus to the full distribution of age gaps within each movie by adding packaging metrics that capture concentration. This allows two movies with similar top gaps to be separated if one is dominated by a single couple and the other has gaps spread across couples.

## 2. EDA & Preprocessing: Updates

In M2 and M3, the movie level features mostly summarized the largest one or two couples. This worked for spotting extreme gaps, but it sometimes grouped together movies that had the same maximum gap for very different reasons. A film can have one unusually large gap with the rest being small, or it can have several couples with meaningful gaps, and the top two statistics do not always separate those cases clearly. That motivated shifting from a top two focus to measures that describe the full within film distribution.

I keep the same EDA, cleaning, and movie table construction from M2 and M3 for comparability, and the only preprocessing update in M4 is adding headline share and HHI as two new columns to the movie level feature table(Figure 1). I chose these because they directly capture within film packaging in a simple and interpretable way. Headline share measures the fraction of the total age gap in a movie that comes from the single largest gap couple, so it is high when one couple dominates the gap profile. HHI measures how concentrated the gaps are across couples, so it increases when gaps are unevenly distributed and is lower when gaps are more evenly spread. Together, they support the refined goal of separating headline driven movies from movies where age gaps recur across multiple couples.

## 3. Analysis & Experiments

**Analysis 1: Feature refinement with packaging metrics**

In this analysis, I refine the movie level features, so they match my main question about within film patterns. Since my goal is to tell apart movies where one couple dominates from movies where gaps show up across multiple couples, it makes more sense to add features that describe the full distribution of gaps within each film.

To do this, I kept the same M3 preprocessing and movie table and only added two new columns, headline share and HHI. Headline share is the largest gap divided by the total gap in the movie, and HHI measures how concentrated the gaps are across couples. I also ran a zero sanity check because a small number of records have age gap equal to zero, and concentration style metrics can sometimes behave oddly when zeros are present. I first measured how common zeros were and checked whether they were concentrated in a few movies, since that would be more likely to cause problems. Then I recomputed headline share and HHI after dropping zero gaps within each movie and compared the updated values to the originals. Nothing changed for any movie, which tells me the zero cases are not affecting(Figure 2) the packaging metrics, so I did not add extra filtering and kept the original feature definitions.

**Analysis 2: KMeans tuning and stability across k and seeds**

In this analysis, I focus on whether the KMeans clusters are reliable or if they change depending on random initialization. This matters in Milestone 4 because the goal is not only to form clusters, but to show that the refined segmentation is strong and repeatable rather than a onetime result. KMeans fits this setting since the movie table is fully numeric and the output is easy to interpret, but it can be sensitive to starting centroids. I expected that if the segments reflect real structure, then repeated runs across different random seeds should produce nearly the same labels for a reasonable range of k values.

To test this, I ran KMeans for k values from 2 to 6 and repeated each k across many random seeds. For each k, I tracked silhouette and inertia as basic quality checks, and I used ARI and NMI to measure how consistent the cluster labels were compared to a reference run. I did this twice, once using the baseline M3 features( Figure 3: Baseline) and once using the updated feature( Figure 3: Updated) set with headline share and HHI. The baseline results were already very stable across seeds, and the updated features stayed extremely stable for k equal 2 to 4, while stability dropped more at k equal 6. One challenge is that silhouette did not point to a single best k, since separation generally decreased as k increased even when stability stayed high, so I did not treat it as the final decision rule. Instead, I used stability plus interpretability to justify a simple k for the rest of the analysis and avoided higher k values where stability began to weaken.

**Analysis 3: Baseline versus updated clustering comparison and interpretation**

In this analysis, I directly test whether adding the M4 packaging features changes the movie segmentation in a meaningful way. I keep k fixed at 3 and fit KMeans twice on the same movie table, once using the baseline M3 features and once using the updated feature set that adds headline share and HHI. This comparison(Figure 4) matters for Milestone 4 because it shows whether the refinement reclassifies movies based on within film concentration, rather than simply reproducing the same clusters with new labels.

- The "Ensemble high gap" group is basically unchanged. Baseline cluster 2 maps cleanly to updated cluster 2, with 25 movies staying in "Ensemble high gap."
- The main change happens in the lower to moderate gap part of the data. Movies that were split between baseline "Low gap typical" and baseline "Headline dominant" get re-sorted into updated "Low gap typical" versus updated "Headline packaged."

The cluster medians support(Figure 5) this interpretation. Updated "Headline packaged" shows very high concentration with median headline share equal to 0.88 and median HHI equal to 0.78, even though its median gap is still moderate at about 7.5. Updated "Low gap typical" has similar overall gap levels but noticeably lower concentration, which is exactly the distinction the packaging metrics were designed to capture.

**Analysis 4: Visualization and couple level validation**

In this analysis, I use plots to make sure the updated clusters are visually supported and interpretable. The PCA plots show that a two component view captures most of the structure in the data, about 79 percent of variance for the baseline features(Figure 6) and about 81 percent for the updated features(Figure 7), so the 2D separation is a useful summary rather than a random projection. In the baseline PCA, the three groups are fairly separated, mostly along overall gap size and how dominant the top couple is. After adding headline share and HHI, the PCA layout shifts and the cluster boundaries reorganize, which lines up with the moderate baseline versus updated agreement and suggests the new features are changing the grouping in a real way. The packaging space plot(Figure 8) makes the refined story easiest to see because it uses the two new metrics directly, and the "Headline packaged" cluster concentrates in the high headline share and high HHI region.

For a quick validation, I map updated movie clusters back to couple level rows and summarize outcomes by cluster(Figure 9). The ordering is consistent with the cluster meanings. "Low gap typical" has the lowest mean gap and lowest Rule of 7 violations, "Headline packaged" is slightly higher, and "Ensemble high gap" is much higher on both, with a strong older man skew. This gives a simple check that the clusters align with meaningful couple level relationship patterns.

## 4. Findings & Interpretations

This milestone makes the movie segmentation easier to understand in real terms. The main finding is that movies are not only different in how large their age gaps are, but also in how those gaps are concentrated within the film. After adding the packaging features, the clusters are separated into three clear story types. There are movies that are genuinely "Low gap typical," movies that are "Headline packaged" where one couple dominates the age gap profile, and movies that are "Ensemble high gap" where large gaps show up more broadly across couples. The packaging space plot (Figure 8) is the clearest evidence for this, because it directly places movies on headline share and concentration, and the "Headline packaged" group sits where both are high, matching the intended meaning of the segment.

A second important takeaway is that the refinement mostly changes how we classify low to moderate gap films rather than changing the extreme cases. The "Ensemble high gap" group stays consistent even after adding packaging features, which suggests that extreme age-gap patterns are strong and easy to detect no matter how features are defined. What changes is the middle. Many movies that previously fell into a general "typical" category get re-sorted into either "Low gap typical" or "Headline packaged" (Figure 4). This is important because it shows that earlier top gap summaries were mixing two different kinds of films. One group is truly low gap overall, while the other is more about how one standout couple drives the movie's gap narrative.

One thing that surprised me is that the updated features made the clustering more interpretable even though simple separation scores did not improve. The silhouette values did not clearly increase and sometimes were slightly lower (Figure 3), but the results became easier to explain because the clusters now align with an intuitive concept of packaging. Another important result is that the clusters also map to meaningful couple-level outcomes. "Ensemble high gap" movies show much higher Rule of 7 violations and a much stronger older man skew than the other groups (Figure 9), which supports that this segment reflects a different relationship norm pattern rather than just a different set of engineered features. Overall, this refinement strengthens confidence that the clusters represent meaningful types of age gap portrayal, and it sets up a clearer way to compare representation patterns across directors, genres, or time periods in later analysis.

## Appendix

**Shared GitHub Repository (Required)**

- **Shared team GitHub repository:** Link
- **Individual work:** Link
- **Colab Link:** Link

**Supplemental Material (Highly Recommended)**

- **Figure 1:** Updated Features

| count_large | second_gap | top_gap_dominance | gaps | headline_share | hhi_gap |
|---|---|---|---|---|---|
| 0 | 0.0 | 2.0 | [2, 0] | 1.000000 | 1.000000 |
| 0 | 5.0 | 5.0 | [10, 5] | 0.666667 | 0.555556 |
| 0 | 0.0 | 8.0 | [8, 0] | 1.000000 | 1.000000 |
| 1 | 14.0 | 14.0 | [28, 14] | 0.666667 | 0.555556 |
| 0 | 0.0 | 2.0 | [2, 0] | 1.000000 | 1.000000 |

- **Figure 2:** 0 age gap sanity check

| | headline_share | hhi_gap | headline_share_no0 | hhi_gap_no0 |
|---|---|---|---|---|
| 0 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |
| 1 | 0.666667 | 0.555556 | 0.666667 | 0.555556 |
| 2 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |
| 3 | 0.666667 | 0.555556 | 0.666667 | 0.555556 |
| 4 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |

- **Figure 3:** KMeans Seed Stability Across k

**Baseline:**

| k | silhouette_mean | silhouette_median | inertia_mean | ARI_mean | ARI_median | NMI_mean | NMI_median |
|---|---|---|---|---|---|---|---|
| 2 | 0.471301 | 0.471301 | 676.833175 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |
| 3 | 0.463567 | 0.463567 | 529.689082 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |
| 4 | 0.410819 | 0.410860 | 406.164065 | 0.973900 | 0.971621 | 0.966087 | 0.965189 |
| 5 | 0.422614 | 0.422828 | 339.720492 | 0.985448 | 0.982930 | 0.975815 | 0.974078 |
| 6 | 0.333788 | 0.332416 | 291.670524 | 0.960931 | 0.956134 | 0.957983 | 0.952063 |

**Updated:**

| k | silhouette_mean | silhouette_median | inertia_mean | ARI_mean | ARI_median | NMI_mean | NMI_median |
|---|---|---|---|---|---|---|---|
| 2 | 0.351358 | 0.351358 | 1094.591610 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |
| 3 | 0.340723 | 0.340723 | 769.197433 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |
| 4 | 0.340734 | 0.340734 | 617.315505 | 1.000000 | 1.000000 | 1.000000 | 1.000000 |
| 5 | 0.343585 | 0.343445 | 492.415134 | 0.979261 | 0.991899 | 0.977468 | 0.986710 |
| 6 | 0.369930 | 0.370912 | 419.293729 | 0.849954 | 0.841663 | 0.899484 | 0.896448 |

- **Figure 4:** Comparing Baseline vs. Updated Feature Clusters (k=3)

```
Baseline vs Updated (k=3)
ARI: 0.399
NMI: 0.485
```

| cluster_base \ cluster_up | 0 | 1 | 2 |
|---|---|---|---|
| 0 | 1 | 21 | 14 |
| 1 | 110 | 53 | 0 |
| 2 | 0 | 0 | 25 |

| | cluster_base | cluster_base_name | cluster_up | cluster_up_name |
|---|---|---|---|---|
| 0 | 1 | Low-gap typical | 1 | Headline-packaged |
| 1 | 1 | Low-gap typical | 0 | Low-gap typical |
| 2 | 1 | Low-gap typical | 1 | Headline-packaged |
| 3 | 0 | Headline-dominant | 2 | Ensemble high-gap |
| 4 | 1 | Low-gap typical | 1 | Headline-packaged |

- **Figure 5:** Cluster Profile Medians

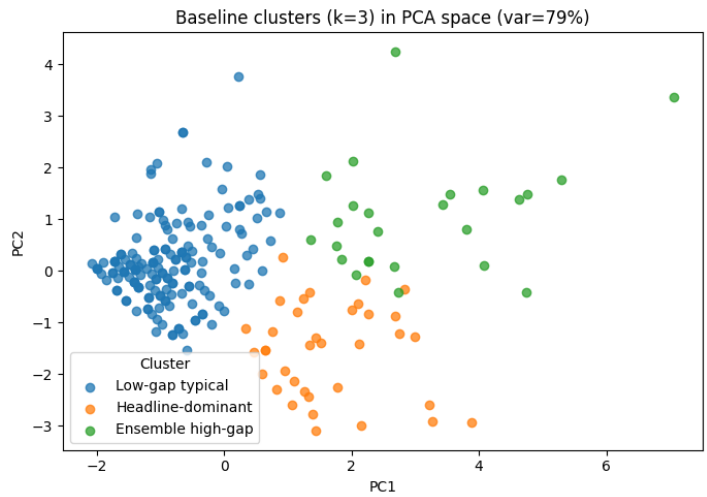
Baseline cluster medians (named):

| cluster_base_name | max_gap | mean_gap | median_gap | second_gap | top_gap_dominance | count_large | n_couples | headline_share | hhi_gap |
|---|---|---|---|---|---|---|---|---|---|
| Ensemble high-gap | 24.0 | 19.5 | 20.00 | 18.0 | 5.0 | 1.0 | 3.0 | 0.47 | 0.39 |
| Headline-dominant | 28.5 | 16.5 | 16.25 | 7.5 | 20.0 | 1.0 | 2.0 | 0.78 | 0.66 |
| Low-gap typical | 10.0 | 7.0 | 6.50 | 4.0 | 4.0 | 0.0 | 2.0 | 0.66 | 0.54 |

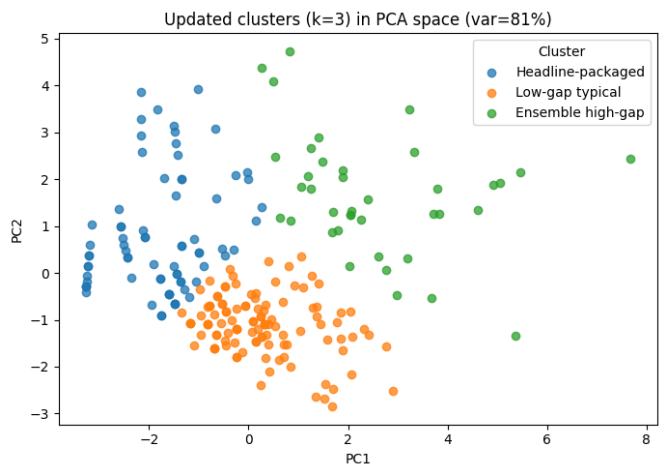
Updated cluster medians (named):

| cluster_up_name | max_gap | mean_gap | median_gap | second_gap | top_gap_dominance | count_large | n_couples | headline_share | hhi_gap |
|---|---|---|---|---|---|---|---|---|---|
| Ensemble high-gap | 26.0 | 20.5 | 20.5 | 16.0 | 9.0 | 1.0 | 3.0 | 0.56 | 0.50 |
| Headline-packaged | 14.0 | 8.0 | 7.5 | 2.0 | 12.0 | 0.0 | 2.0 | 0.88 | 0.78 |
| Low-gap typical | 10.0 | 7.0 | 7.0 | 6.0 | 3.0 | 0.0 | 2.0 | 0.57 | 0.51 |

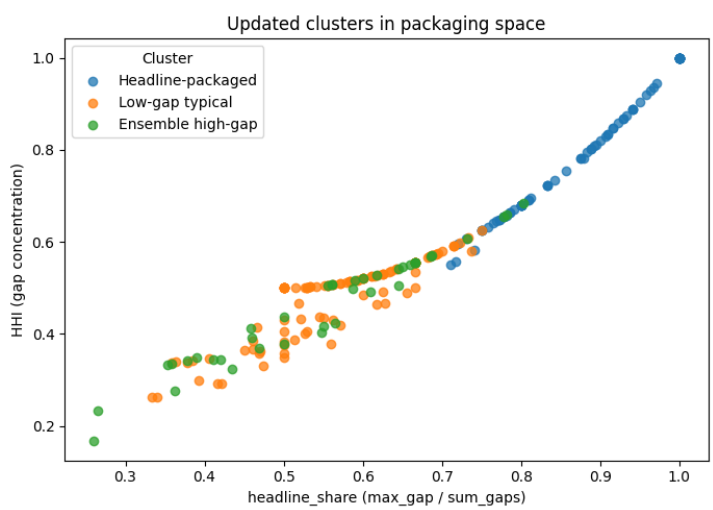
- **Figure 6:**




- **Figure 7:**




- **Figure 8:**



- **Figure 9:** Movie level validation

| | cluster_up_name | n_couples | pct_rule_violation | pct_older_man | mean_gap |
|---|---|---|---|---|---|
| 0 | Low-gap typical | 281 | 18.1 | 73.0 | 7.35 |
| 1 | Headline-packaged | 153 | 24.2 | 75.8 | 8.85 |
| 2 | Ensemble high-gap | 107 | 58.9 | 91.6 | 19.58 |

**Process Overview**

I began by loading the Hollywood Age Gap dataset and applying the same baseline cleaning steps used in M2 and M3 to keep results comparable. I then created a movie level analysis table by aggregating couples within each film, keeping only movies with at least two couples and generating the baseline summary features used previously. For Milestone 4, I refined the movie representation by computing two additional packaging metrics from each movie's full list of couple age gaps, headline share and HHI, and merged them into the movie table. Next, I ran KMeans clustering on both the baseline and updated feature sets, tested stability across random seeds and a range of k values, and selected an interpretable k for the final segmentation. I compared baseline versus updated cluster assignments, profiled clusters using median feature tables, and used PCA and a packaging space scatterplot to visually support the segment meanings. Finally, I validated the movie clusters by mapping them back to couple level rows and summarizing outcomes such as Rule of 7 violations, directionality, and average age gaps to connect the segments to real relationship patterns.

**Use of Generative AI Tools**

I used **ChatGPT** as a writing, structuring, and brainstorming assistant to refine wording, reduce sections to required page limits, and organize the narrative. I also used ChatGPT to help me understand technical concepts, explore solution ideas, and get guidance while debugging or improving code (e.g., interpreting errors, suggesting approaches, and checking syntax). ChatGPT was additionally used to assist with **Markdown formatting** and presentation. All core analytical design, feature engineering, clustering, integration logic, implementation decisions, and interpretation were developed and completed by our team. All AI-assisted output was reviewed, edited, and validated against our actual code, results, and project requirements.

ChatGPT Link: Link